CADETX TEAM PROJECT

MONTH 1 — WEEK 2: CUSTOMER RULES

Project: Heavy Suppliers Warehouse
Prepared by: Rofiat
Assigned Area: Customer and Sales Analysis
Phase: Data Foundation

1. Purpose

The purpose of this document is to define customer rules for analysing purchasing behaviour, identifying repeat customers, recognising high-value customers, and understanding customer churn.

2. Customer Churn Rule

A customer will be considered at risk of churn or inactive when they have not made a purchase within an agreed period.

- **Data columns:** `customer_id`, `last_purchase_date`
- **Rule status:** Proposed; the inactivity threshold requires validation using purchasing patterns and feedback from the coach.
- **Business purpose:** To identify inactive customers and support customer retention strategies.

3. Repeat Customer Rule

A repeat customer is a customer who has placed more than one qualifying sales order.

- **Data columns:** `customer_id`, `so_id`, `order_status`
- **Rule status:** Defined; the treatment of cancelled orders must follow the agreed order-status rules.
- **Business purpose:** To identify returning customers and understand customer loyalty.

4. High-Value Customer Rule

High-value customers are customers who contribute relatively high purchase value to the business.

- **Data columns:** `customer_id`, `total_purchase_value`
- **Rule status:** Proposed; the classification threshold will be determined after reviewing the distribution of customer purchase values.
- **Business purpose:** To identify customers who contribute significantly to sales and may benefit from retention strategies.

5. Purchase Frequency Rule

Purchase frequency measures how often each customer places qualifying sales orders within a specified period.

- **Data columns:** `customer_id`, `so_id`, `order_date`, `order_status`
- **Rule status:** Defined; the analysis period and qualifying order statuses will be confirmed.
- **Business purpose:** To understand customer purchasing habits and identify frequent and less-frequent customers.

6. Data Considerations

The following checks will support accurate customer analysis:

- Use `customer_id` to identify individual customers.
- Use `last_purchase_date` to assess how recently each customer purchased.
- Use `order_date` to analyse purchasing patterns over time.
- Confirm the treatment of cancelled and other excluded orders.
- Review `total_purchase_value` before establishing a high-value customer threshold.
- Validate the churn and high-value customer thresholds with the coach before finalising them.

7. Python Setup

Status: Completed.

The Python environment has been set up and is ready for the next stage of analysis. Further work will involve preparing the data and applying Python to customer analysis and forecasting tasks.

8. Expected Outcome

- Customer analysis rules documented.
- Proposed thresholds identified for validation.
- Python environment set up.
- Outstanding decisions clearly identified for review by the coach.

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